

Designing of Surrogate models for Electromagnetic Compatibility of Power electronic circuits

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CERTIFICATE

This is to certify that the project titled **Designing of Surrogate models for Electromagnetic Compatibility of Power electronic circuits** is a record of the bonafide work done by **Ayush Shukla (200907282)** submitted in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology (BTech) in **ELECTRONICS AND COMMUNICATION ENGINEERING** of Manipal Institute of Technology, Manipal, Karnataka, (A Constituent Unit of Manipal Academy of Higher Education), during the academic year 2023 - 2024.

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With best regards



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ABSTRACT

Electromagnetic Compatibility (EMC) is vital for maintaining signal integrity and preventing interference in electronic systems. It encompasses the study of electromagnetic phenomena to ensure devices coexist without disruption. Compliance with EMC standards safeguards against malfunctions, data corruption, and safety hazards, crucial in sectors like aerospace, medical, and telecommunications. Neglecting EMC considerations can compromise system reliability and pose risks to both equipment and users.

Various Surrogate modelling techniques tasked with exploration and comparison in Electromagnetic Compatibility (EMC) were utilized to expedite complex simulations and optimization processes. These models, often simpler and faster than detailed electromagnetic simulations, approximate the behaviour of intricate systems. By employing techniques such as neural networks, Gaussian Process modelling, polynomial regression, electromagnetic responses to varying conditions are efficiently predicted by surrogate models.

This approach enables the rapid assessment of EMC performance, robust system design, and exploration of large design spaces without the computational burden of full simulations, thereby accelerating product development and ensuring compliance with EMC standards. Additionally, the tasks involved the utilization of OptisLang, a tool provided by ANSYS, to study and compare the models available and those provided by OptisLang. Along with this, various error validation methods such as Feature Selective Validation were explored and implemented for various usecases. Soon, after a near-perfect modelling technique that is accurate and fast enough has been found, circuits will be optimized as per specified EMI/EMC regulations.

Some Tools and Technologies that are used are Python , UQLab , OptisLang , LTSpice

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CHAPTER 1

INTRODUCTION

This Chapter talks about the main area of work, i.e. Electromagnetic Interference and Electromagnetic Compatibility and how it has come into the picture in the modern-day scenario with some interesting examples. Then, the motivations behind the work are introduced. These include why and by whom EMC is required on a business scale, how the approach taken here is unique, and the possible end results of the whole project.

1.1 About EMI and EMC:

Electromagnetic Interference (EMI) refers to the disturbance caused by electromagnetic radiation from an external source that can disrupt the performance of electronic systems or devices. This interference can be natural, like the disturbances caused by solar flares that can sometimes be observed in analogue TV signals. However, most EMI is human made, encompassing everything from power lines to handheld devices. This is because electronic components inherently emit some radiation when powered by a current.

Electromagnetic Compatibility (EMC) deals with ensuring that devices can operate within their electromagnetic environment without causing or experiencing interference. The world of EMC has become increasingly important with the widespread adoption of wireless technologies, IoT devices, and electric vehicles. In practice, EMC involves designing electronic systems to minimize their own EMI emissions while maximizing their electromagnetic susceptibility, which refers to a device's ability to resist external EMI. This delicate balance ensures harmonious coexistence for all electronic devices within our increasingly electromagnetic world.

1.2 Area of work:

Research is being conducted at the Christian Doppler Laboratory, Institute of Electronics, TU Graz, led by Dr. Jan Hansen, to address EMI/EMC problems encountered in electric vehicles (EVs).

Electric vehicles integrate numerous electrical components, including batteries, power converter circuits, and radio antennas. Due to space limitations within the vehicle, these components are densely packed, leading to EMI issues as described previously. Traditional simulation and optimization procedures are not feasible for such complex circuits due to the extensive human hours and computational resources required. Running the sheer number of simulations necessary would be impractical.

Therefore, the research leverages various mathematical modelling techniques and machine learning methods. These approaches offer significant advantages in terms of speed and computational efficiency compared to traditional electromagnetic simulation tools such as Ansys HFSS and CST Studio.

1.3 Motivation for the work:

Due to the relative novelty of Electromagnetic Compatibility (EMC) within electrical engineering, a limited amount of research has been conducted in this domain. However, as electric vehicle technology advances, ensuring EMC becomes crucial for the long-term sustainability of these electrical systems. Recent breakthroughs in machine learning and the increasing computational power of computers offer promising avenues to address this challenge. By conducting research in this area, significant progress can be made towards solving EMI/EMC problems in electric vehicles.

Participation in this project fosters valuable interactions with highly respected individuals within the scientific community. This collaborative environment provides an exceptional learning opportunity, allowing the researcher to gain significant knowledge from the experiences of these experts. This continuous learning process not only facilitates exploration of new ideas but also aids the researcher in charting their future career path. Furthermore, the project contributes to the development of the researcher's technical skillset in both electrical and electronics engineering and the ever-evolving field of machine learning. Machine learning, with its broad applicability across various scientific disciplines, offers a particularly valuable skill to cultivate.

The project's methodology distinguishes itself by combining machine learning and mathematical modelling techniques with the field of electromagnetics. This unique approach creates a stimulating research environment brimming with opportunities for intellectual growth and discovery.

1.4 Objective of the work:

The main objectives of the work were:

- Designing, optimising and implementing Surrogate models for various Circuits and systems
- Using the surrogate models, improve the circuits and systems' Electromagnetic Compatibility

1.5 Target Specifications:

Surrogate modelling in EMC is important for two key reasons:

- **Efficiency in Design and Optimization:** Surrogate models provide a faster and more computationally efficient way to predict the behaviour of complex EMC systems compared to full-scale simulations.
- **Enhanced Understanding and Prediction:** Surrogate models can help in gaining insights into the influence of various factors on EMC performance. By providing accurate approximations of system behaviour, they enable better prediction and management of electromagnetic interference, leading to more robust and reliable electronic designs.

1.6 Project work schedule:

Table 1.1 Project work schedule

February	Reviewing previously done work, Literature survey
March	Analysis of Kriging and other Surrogate models and improving them
April	Benchmarking Ansys OptisLang and comparison with other surrogate models
May	Exploring and Implementing Feature Selective Validation (FSV) for various testcases and its further improvement and validation
June	Further validation of FSV and documentation

CHAPTER 2

BACKGROUND THEORY

2.1 Introduction to the Project title:

“Designing of Surrogate models for Electromagnetic Compatibility of Power electronic circuits” – This research investigates the development and application of surrogate models for ensuring electromagnetic compatibility (EMC) in power electronic circuits. The focus is on understanding, designing, optimizing, and comparing various modelling techniques. These techniques include those developed by collaborators within the research group, by other engineers and researchers internationally, and those provided by industry.

By employing these techniques, surrogate models are constructed to represent the behaviour of the circuits. These models are then used to optimize the circuit design to comply with established industry EMC regulations, such as CISPR 25, which defines the limits for conducted and radiated emissions of information technology equipment. This optimization process allows for the development of circuits that minimize electromagnetic interference while maintaining functionality, ultimately contributing to a more harmonious electromagnetic environment.

2.2 Literature Review:

In the past, EMC modelling techniques were primarily based on analytical methods and empirical formulas. These approaches provided basic insights into electromagnetic phenomena but had limitations in accurately predicting complex interactions in real-world systems. Simplified circuit models, transmission line models, and basic electromagnetic theory were commonly used to analyse EMC issues. [1]

Due to the development of computational electromagnetics and numerical simulation methods, significant advancements have been made in EMC modelling techniques. Finite element analysis (FEA), finite-difference time-domain (FDTD), method of moments (MoM), and other numerical techniques are now extensively used to model electromagnetic fields and interactions in complex structures and environments [2]. These methods offer higher accuracy and versatility, enabling a more detailed and efficient analysis of EMC problems for engineers. Additionally, machine learning and data-driven approaches are being increasingly integrated into EMC modelling to enhance prediction capabilities and automate design processes [6].

The future of EMC modelling techniques is expected to involve further advancements in computational methods, simulation algorithms, and interdisciplinary approaches. There will be

a continued focus on improving the accuracy, efficiency, and scalability of EMC models. This is necessary to address the growing complexity of electronic systems and the expanding electromagnetic spectrum. Integration of Multiphysics simulations, which incorporate thermal, mechanical, and electromagnetic effects, will become more prevalent to provide comprehensive analyses of EMC issues. Additionally, advancements in artificial intelligence and quantum computing may pave the way for new avenues in modelling and simulating electromagnetic phenomena. These advancements could achieve unprecedented accuracy and speed.

2.3 Research Gap:

Surrogate modelling has emerged as a powerful tool in the field of Electromagnetic Compatibility (EMC) due to the computational demands of simulating complex electronic systems [3]. Traditional full-wave simulations, while providing high accuracy, can be time-consuming, especially when performing optimization tasks or exploring design variations. Surrogate models, on the other hand, offer a faster and more efficient alternative by learning the relationship between design parameters and EMC performance from a limited set of simulations.

Several surrogate modelling techniques have been explored for EMC applications. Kriging, a popular choice, leverages Gaussian processes to create a statistical representation of the relationship between design parameters and emissions [7]. This allows for uncertainty quantification alongside the predicted response, providing valuable insights into the reliability of the prediction. Other techniques like polynomial chaos expansions and artificial neural networks have also shown promise in specific scenarios [2]. Based on the intended level of accuracy, the computational resources available, and the system complexity, a surrogate model is selected. By employing surrogate models, we can significantly reduce simulation times, enabling faster design iterations, optimization of filter components, and robust design against EMI for various operating conditions [8].

Another area of Interest was to find an appropriate error validation method. In computational electromagnetics, validation is a formal procedure that verifies the model's or the modelling technique's expected behaviour. One data set can be subtracted from the other to see the differences, but this is not a reliable way to determine whether the two sets are generally comparable or in agreement. [15].

2.4 Theoretical Discussions:

Electromagnetic Compatibility (EMC) ensures electronic systems function properly in their intended environment without causing or being unduly influenced by electromagnetic interference (EMI). Modelling plays a crucial role in achieving this goal by enabling:

- **Prediction of EMI emissions:** Modelling allows predicting the electromagnetic fields generated by a system under various operating conditions. This helps identify potential emission issues before physical prototypes are built.
- **Analysis of susceptibility to EMI:** Modelling can be used to assess how susceptible a system is to external electromagnetic fields. This helps determine if the system will malfunction in real-world environments.
- **Design optimization for EMC:** By simulating the effects of design changes on EMI, engineers can optimize system components and layouts to minimize emissions and enhance immunity.

Several theoretical frameworks underpin modelling techniques used in EMC:

- **Maxwell's Equations:** These fundamental equations govern the interaction between electric and magnetic fields. Numerical methods like Finite Element Method (FEM) and Finite Difference Time Domain (FDTD) solve Maxwell's equations for complex geometries, providing a highly accurate representation of electromagnetic behaviour.
- **Circuit Theory:** This framework allows modelling the electrical behaviour of circuits using components like resistors, capacitors, and inductors. Circuit simulations can predict conducted emissions and susceptibility through cables and power lines.
- **Transmission Line Theory:** This theory models the transmission of electromagnetic signals along conductors, considering factors like impedance matching and reflections. It helps design filters and transmission lines to attenuate unwanted emissions.
- **Statistical Methods:** Statistical models can be used to characterize the randomness and uncertainty associated with EMI phenomena. Techniques like Monte Carlo simulations can be used to account for variations in manufacturing tolerances and environmental conditions.

Choosing the Right Modelling Approach: The choice of appropriate modelling technique depends on several factors:

- Complexity of the System: Simple circuits may be modelled effectively using circuit theory, while complex systems with intricate geometries require full-wave simulations.
- Desired Level of Accuracy: High-fidelity simulations may be necessary for critical applications, while faster approximate methods may suffice for initial design exploration.
- Computational Resources: Full-wave simulations can be computationally expensive. Surrogate modelling techniques like Kriging can be used to build faster approximations based on a limited set of simulations.

Advantages and Limitations of Modelling:

- Benefits: Enables faster design iterations, reduces hardware prototyping costs, and facilitates optimization for robust EMC performance.
- Limitations: Accuracy depends on the chosen model and the underlying assumptions.

Validation with experimental measurements is crucial for real-world application.

In conclusion, theoretical frameworks based on Maxwell's equations, circuit theory, and statistical methods provide the foundation for various modelling techniques used in EMC. By carefully selecting and applying these methods, engineers can achieve efficient and effective design practices for ensuring electromagnetic compatibility.

2.5 Mathematical Derivations:

Surrogate modelling in EMC leverages mathematical techniques to create approximate models of complex relationships between design parameters and EMC performance metrics (e.g., emissions, susceptibility). While the underlying full-wave simulations might involve solving Maxwell's equations, I can briefly discuss about the mathematics behind a Modelling technique that I studied and made use of: Kriging.

Kriging - also known as Gaussian Process Regression, is a powerful surrogate modelling technique based on the assumption that the data points representing design parameters and corresponding EMC performance metrics are samples from a continuous Gaussian Process (GP). This GP (equation 1) defines a statistical relationship between the input (design parameters) and output (EMC performance) variables [6].

$$\psi^{(i)} = \exp\left(-\sum_{j=i}^k \theta_j |x_j^i - x_j|^{\rho_j}\right) \quad (1)$$

In order to construct a Kriging model, we first begin with a set of sample data , $\mathbf{X} = \{\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(n)}\}^T$, and observed responses $\mathbf{Y} = \{\mathbf{y}^{(1)}, \mathbf{y}^{(2)}, \dots, \mathbf{y}^{(n)}\}^T$ and we next seek to create an expression for a predicted value at a new point $\mathbf{x}$.

We need to start with a very abstract idea: we're going to treat our observed reactions as though they constitute an arbitrary procedure. Equation 2 illustrates the set of random vectors that we use to represent this.

$$\mathbf{Y} = \begin{pmatrix} Y(x^1) \\ \vdots \\ Y(x^n) \end{pmatrix} \quad (2)$$

This random field has a mean of $\mathbf{1}\mu$ ($\mathbf{1}$ is an $n \times 1$ column vector of ones). The random variables are correlated with each other using the basis function expression:

$$\text{Cor}[Y(x^i), Y(x^l)] = \exp\left(-\sum_{j=1}^k \theta_j |x_j^i - x_j^l|^{p_j}\right) \quad (3)$$

From this we can create an $n \times n$ correlation matrix (equation 4) , Ψ , with all the observed data:

$$\Psi = \begin{vmatrix} \text{cor}[Y(x^1), Y(x^1)] & * & * & * & \text{cor}[Y(x^1), Y(x^n)] \\ * & & & & * \\ * & & & & * \\ \text{cor}[Y(x^n), Y(x^1)] & * & * & * & \text{cor}[Y(x^n), Y(x^n)] \end{vmatrix} \quad (4)$$

We also get a covariance matrix $\text{Cov}(\mathbf{Y}, \mathbf{Y}) = \sigma^2 \Psi$

This assumed correlation between the sample data shows our expectation that an engineering function will behave in a certain way – importantly being smooth and continuous throughout the range. The matrix Ψ describes our current set of random variables ($\mathbf{Y}$), which are correlated in some manner. The parameters p_j and θ_j , as well as the absolute distance between the sample points $|x_j^i - x_j^l|$, determine these correlations.

The parameter ‘p’ is basically the smoothness parameters, reducing p_j increases the rate at which the correlation initially drops as $|x_j^{(i)} - x_j|$ increases.

In essence, θ_j is the breadth parameter that determines the extent to which the impact of a sample point extends. A high θ_j indicates a large difference between the θ_j of $Y(x_j)$ and all other points in our sample, but a low θ_j indicates that all points will have a greater correlation and $Y(x_j)$ will be more comparable.

Now we understand what the correlation means, but need to decide on the value of p and θ . One methodology is to choose p and θ to maximise the likelihood of $\mathbf{y}$.

- Our likelihood $\mathbf{y}$ is:
$$L(Y^{(1)} \dots Y^{(n)} | u, \sigma) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp \left[\frac{-\sum (Y^i - u)^2}{2\sigma^2} \right] \quad (5)$$

Which can be expressed in terms of sample data as:

$$L = \frac{1}{(2\pi\sigma^2)^{n/2} |\psi^{1/2}|} \exp \left[\frac{-(y-1u)^T \psi^{-1} (y-1u)}{2\sigma^2} \right] \quad (6)$$

To simplify the likelihood maximization, we take the natural logarithm and get:

$$\ln(L) = \frac{-n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2} \ln|\psi| - \frac{(y-1u)^T \psi^{-1} (y-1u)}{2\sigma^2} \quad (7)$$

Now, we take derivative of equation 7 and set it to zero, we obtain maximum likelihood estimates (MLEs) for σ^2 and μ

$$\hat{\mu} = \frac{1^T \psi^{-1} y}{1^T \psi^{-1} 1} \quad \widehat{\sigma^2} = \frac{(y-1u)^T \psi^{-1} (y-1u)}{\eta} \quad (8)$$

These MLE can be substituted now back into the equation 7, and then when the constant terms are removed, we get that is known as the ***concentrated ln-likelihood function***

$$\ln(L) \approx -\frac{n}{2} \ln \widehat{\sigma^2} - \frac{1}{2} \ln|\psi| \quad (9)$$

After some more mathematical processes (quite complex to explain it here), we obtain the final MLE:

$$\bar{y} = u + \psi^T \Psi^{-1} (y - 1u) \quad (10)$$

2.6 About Pareto solutions:

Optimizing EMC involves balancing conflicting goals like minimizing emissions and keeping costs low. Pareto solutions offer a set of designs that represent the best possible trade-offs between these objectives. By analysing this Pareto front, we can make informed decisions and achieve optimal EMC performance within project constraints [14].

Pareto front refers to the collection of all Pareto solutions for a given optimisation problem. This front visually depicts the achievable trade-offs between the objectives, often in a plot or a visual form.

2.7 About Feature Selective Validation:

Validation in Computational Electromagnetics is a formal process that ensures that the model or modelling technique behaves in an expected manner [19]. While it can demonstrate differences, a straightforward subtraction of one data set from the other will not provide a reliable picture of overall agreement or similarity.

The FSV method provides a more comprehensive picture of the agreement by combining a feature-based comparison with an amplitude-based comparison. For relatively subjective decisions, the FSV has been calibrated to mirror human "expert" comparison; nonetheless, we apply natural language labels like "excellent," "very good," "good," "poor," etc. Regarding the analogies [16].

The FSV technique is based on splitting the data to be compared into two "component" measures and then recombining the results to get a measure of overall goodness of fit. While the Feature Difference Measure (FDM) looks at the features that fluctuate quickly (as a function of the independent variable), the Amplitude Difference Measure (ADM) compares the amplitudes and "trends" of the two data sets. [17, 18]. Subsequently, the ADM and FDM are integrated to provide a Global Difference Measure (GDM), which provides a comprehensive quantitative comparison between the datasets.

Reporting the Global Difference Measure (GDM) single number is the minimal expectation.

2.8 Conclusions:

In conclusion, the background theory section provides a comprehensive understanding of the key concepts and principles essential for the design and optimization of surrogate models for Electromagnetic Compatibility (EMC).

It outlines the significance of EMC in ensuring the reliable operation of electronic systems amidst electromagnetic interference. By delving into fundamental theories such as electromagnetic field theory, signal integrity, and statistical methods, this report lays the groundwork for developing effective surrogate models.

Moreover, it highlights the critical role of surrogate models in accelerating the EMC design process, reducing costs, and enhancing product performance. Through the synthesis of theoretical frameworks and practical applications, this section sets the stage for the subsequent stages of research and experimentation aimed at advancing EMC optimization techniques. The section also talks about Feature Selective Validation, an error validation technique and how it has been ad

CHAPTER 3

METHODOLOGY

3.1 Introduction:

This section talks about investigating surrogate models for achieving EMC in power electronic circuits. It involves understanding EMI generation and existing lab models. We explored faster mathematical models implementable with various Software and Tools. Analysis focused on the filter and DC-DC converter circuits, with emphasis on developing a suitable model for the latter. Validation techniques like K-Fold Cross-Validation ensured model generalizability. Finally, comparison and refinement of various models like Wide Band Kriging [7] and Deep Infinite Mixture Gaussian Process for optimal efficiency and accuracy are discussed.

3.2 Methodology:

1. Research on Electromagnetic Interference and Electromagnetic Compatibility:

Every component in an electric circuit has the potential to contribute to EMI. Here are some common culprits:

- High-speed switching circuits: These circuits generate rapid changes in current, which can induce voltage spikes and radiate electromagnetic waves.
- Long wires and cables: These act like antennas, both picking up and radiating EMI. We can imagine a long, loose antenna picking up radio signals.
- Poor grounding: Improper grounding can create unwanted current paths that can cause EMI issues.

EMI can be broadly categorized into two types based on its mode of transmission:

- Conducted Emission: This occurs when EMI travels along wires and cables connected to the circuit.
- Radiated Emission: This occurs when EMI travels through the air as electromagnetic waves. It's like the circuit giving out electrical noise into the world.

Conducted EMI can be further classified into two modes depending on the reference path of the interfering current:

- Common Mode Emission: This occurs when the EMI current flows in the same direction through all the conductors in a cable (basically all the wires in a cable carrying the EMI current together).
- Differential Mode Emission: This occurs when the EMI current flows in opposite directions through the conductors in a balanced cable (we can think it as the EMI current flowing out on one wire and returning on another)

Several techniques can be employed to minimize EMI and achieve EMC regulations:

- Circuit design: Careful circuit layout, component selection, and proper grounding practices can significantly reduce EMI generation.
- Shielding: Enclosing the circuit in a metal enclosure can block radiated EMI.
- Filtering: Filters can be used to block unwanted frequencies from entering or leaving the circuit.

2. Improving on the existing work

A highly accurate model for predicting electromagnetic radiation in specific circuits, such as filter and DC-DC converter circuits, had already been developed by previous researchers at the laboratory. This research builds upon that foundation by focusing on a deeper understanding of these circuits. My investigation explored how individual components influence EMI generation and potential mitigation strategies. Additionally, the research is tasked with identifying alternative approaches to existing models. The focus is on exploring faster and computationally less demanding mathematical models or techniques suitable for implementation with the technology in our hands.

3. Analysing the Circuits and the models

The first circuit shown is a Filter circuit in which we were measuring how the EMI is affected by changing the values of the components of the circuit , such as the capacitance, inductance and the resistance as shown in Table 1. We wish to vary the values of the components to attain a Circuit which obeys the EMC regulations.

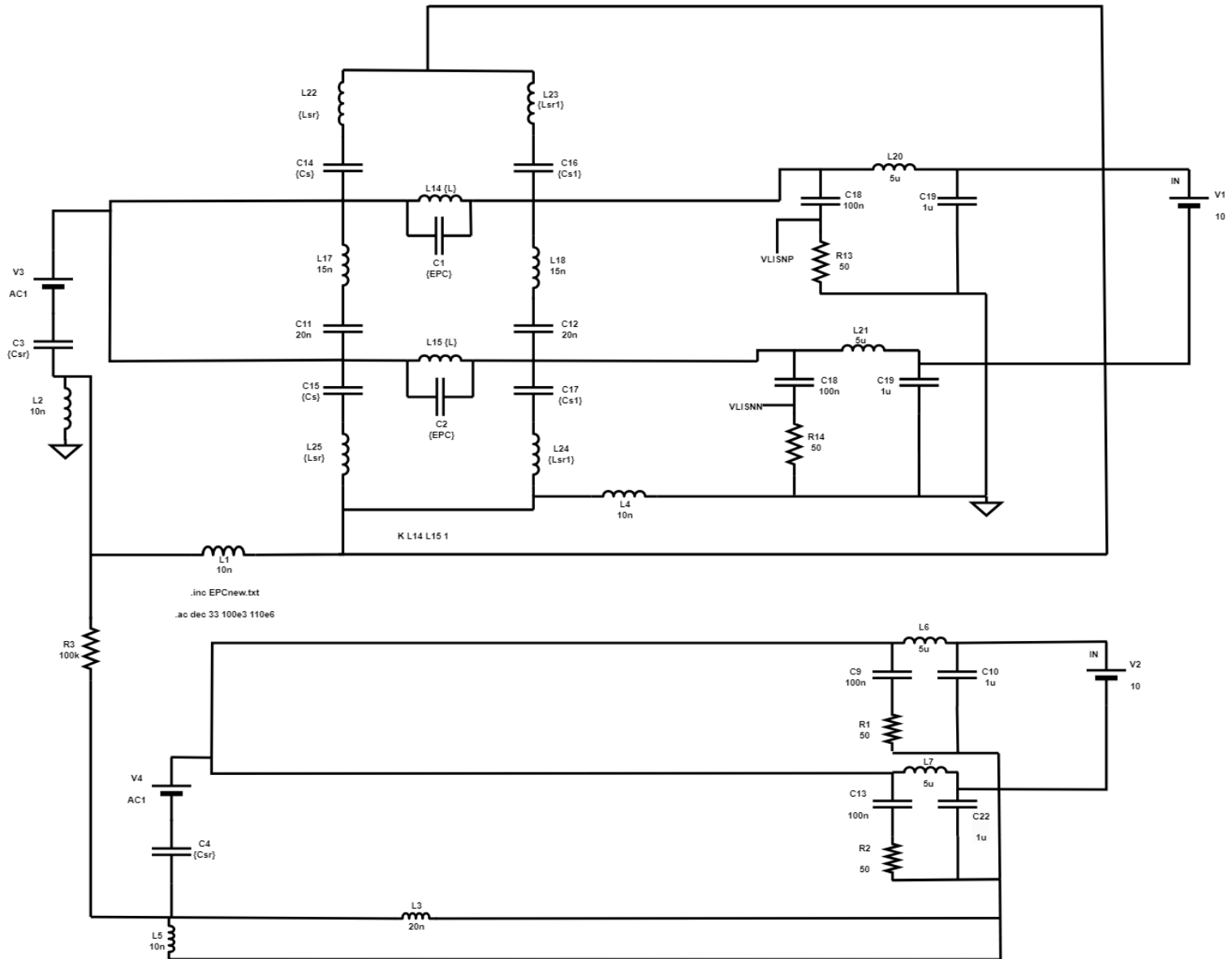


Figure. 3.1 filter circuit with 7 varying components

There are the range of values as shown :

Table 3.1 Design Parameter Ranges

Name	Range	Function
L_{CM}	$1 \mu H - 50 \mu H$	CM choke inductance
C_{CM}	$0.2 pF - 200 pF$	CM parasitic capacitance
C_{LCM}	$0.1 pF - 50 pF$	Parallel capacitance of CM choke inductance
C_{yin}	$2.2 nF - 100 nF$	Cy capacitance – Input side
L_{Cyin}	$4.7 nH - 25 nH$	Cyin parasitic inductance
C_{yout}	$22 nF - 1000 nF$	Cy capacitance-Output side
L_{Cyout}	$4.7 nH - 25 nH$	Cyout parasitic inductance

The second circuit I worked on modelling is a DC-DC Half Bridge converter circuit as shown

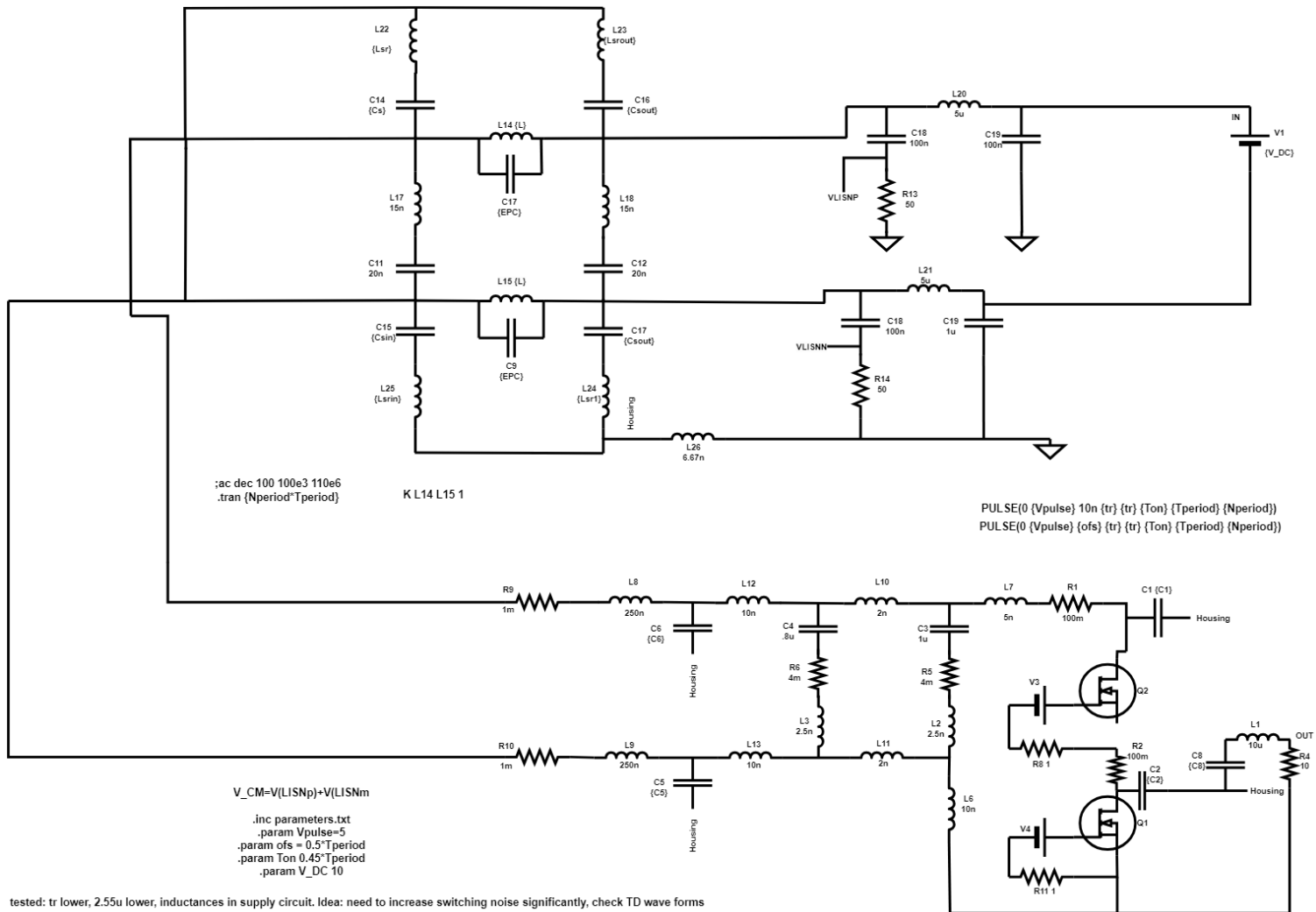


Figure. 3.2 Half bridge circuit with 13 varying components

The half-bridge converter is a popular DC-DC converter circuit using two switches in a half-bridge configuration. It efficiently steps up or down voltage by controlling the switching of the input voltage across an inductor, resulting in a regulated output voltage. This simple design makes it versatile for various applications like power supplies and solar inverters. The schematic also consists of an EMI Filter.

Table 3.2 Design Parameter Ranges for DC-DC Half bridge converter

Name	Range	Function
L_{CM}	$1 \mu H - 50 \mu H$	CM choke inductance
C_{LCM}	$0.1 pF - 50 pF$	Parallel capacitance of CM choke inductance
C_{yin}	$2.2 nF - 100 nF$	Cy capacitance – Input side
L_{Cyin}	$4.7 nH - 25 nH$	Cyin parasitic inductance
C_{yout}	$22 nF - 1000 nF$	Cy capacitance-Output side
L_{Cyout}	$4.7 nH - 25 nH$	Cyout parasitic inductance
C1	1-50 pF	Capacitance
C2	1-50 pF	Capacitance
C5	1-50 pF	Capacitance
C6	1-50 pF	Capacitance
C8	1-50 pF	Capacitance
tr	1-30 n seconds	Rise time
tp	5-10 u seconds	Pulse width

We wish to vary the values of the components to attain a Circuit which obeys the EMC regulations. Here we have more components to deal with and so the complexity of the circuit increases and so does the importance of accurate prediction. The work on modelling this circuit is in the works.

4. Exploring about some other Validation methods (Feature Selective Validation)

In the world of EMC (Electromagnetic Compatibility), surrogate models have emerged as powerful tools for faster and more efficient analysis. However, just like any model, their predictions need to be validated to ensure they provide reliable results. Some of the methods explored were:

1. K-Fold Cross-Validation:

- Using this method, the available data is divided into k folds, such as 10 folds. After training on k-1 folds, the model is tested on the remaining fold. This procedure is carried out k times, each time utilising a different fold for testing
- The average error over all k folds is then used to assess the model's overall performance. By using this technique, one can determine how generalizable the model is and prevent overfitting to the training set.

2. Leave-One-Out Cross-Validation (LOOCV):

- The number of data points, denoted by k , is the specific case of k -fold cross-validation in this instance. The remaining data is used for training, and each data point is only used once for testing.
- Although LOOCV offers a thorough examination, it might be costly to compute for large datasets.

3. Feature Selective Validation (FSV)

FSV analyses data more smartly: it looks at overall shape and key features [10]. It breaks down two datasets into:

- i. Overall similarity score: A single number for general agreement, valid across the whole range
- ii. Point-by-point comparison: Shows where data matches or differs.

A successful implementation of FSV in Python was achieved.

The Algorithm of FSV [15] is as follows:

a) Generation of the working datasets:

- Ensure that the datasets to be compared overlap in a region.
- Identify the number of datapoints.
- If need be, apply interpolation to ensure the datapoints of the two datasets are coincident.

b) Fourier Transform both datasets.

c) The following procedure to be done on both datasets:

- Take the first four data points in the modified data set and apply an inverse Fourier transform to their contents. The returning vectors will have the labels $DC1(n)$ and $DC2(n)$, respectively.
- Beginning at the fifth data point and working your way through the entire data set, determine the sum of the values, or S , of the independent variable. Identify the position of the "40%." This is the data point where the total of the independent variable's values, starting with the fifth data point and going all the way up to the data point, equals or exceeds 40% of the computed value.

$$S = \sum_{i=5}^N TWDS(i) \quad (11)$$

Where TWDS(i) is the i^{th} independent element in the converted dataset, "i" is the dataset element, N is the total length of the dataset, and S is the sum of the independent variables.

- Establish the position of the "breakpoint." This is five data points more than the lowest of the two "40% location" values found in the previous step.

$$I_{bp} = I_{40\%} + 5 \quad (12)$$

- To create the filtered converted working data set, to the independent variable values contained in the designated elements of the modified working data set, apply the filter indicated in Table 3.3.

Table 3.3 Filter table a

Element number	Filter value
$I_{bp} - 3$	1.000
$I_{bp} - 2$	0.834
$I_{bp} - 1$	0.667
I_{bp}	0.500
$I_{bp} + 1$	0.334
$I_{bp} + 2$	0.167
$I_{bp} + 3$	0.000

- On the filtered converted working data set, apply an inverse Fourier transform. The vectors that Lo1(n) and Lo2(n) are to be returned.
- To create the filtered converted working data set, apply the filter shown in Table 3.4 to the independent variable values found in the designated elements of the transformed working data set.

Table 3.4 Filter table b

Element number	Filter value
$I_{bp} - 3$	0.000
$I_{bp} - 2$	0.167
$I_{bp} - 1$	0.334
I_{bp}	0.500
$I_{bp} + 1$	0.667
$I_{bp} + 2$	0.834
$I_{bp} + 3$	1.000

- Perform an inverse Fourier transform on the filtered transformed working data set. The vectors to be returned as $Hi_1(n)$ and $Hi_2(n)$
- d) For each set of datapoints within the working datasets, the ADM to be computed using the equation (13)

$$ADM(n) = \left| \frac{\alpha}{\beta} \right| + \left| \frac{x}{\delta} \right| \exp \left\{ \left| \frac{x}{\delta} \right| \right\} \quad (13)$$

where:

$$\alpha = (|Lo_1(n)| - |Lo_2(n)|)$$

$$\beta = 1/N * \sum_{i=1}^N (Lo_1(i) + Lo_2(i))$$

$$X = (|DC_1(n)| - |DC_2(n)|)$$

$$\delta = 1/N * \sum_{i=1}^N (DC_1(i) + DC_2(i))$$

- e) Now Calculate the mean value of the ADM from equation (14).

$$ADM = \sum_{i=1}^N ADM(i) / N \quad (14)$$

- f) The next step is to construct a confidence histogram of the ADM. The confidence histogram offers some inference to assign a single figure of merit to the comparison and functions similarly to a probability distribution function.

Table 3.5 FSV interpretation scale

FSV value (quantitative)	FSV interpretation (qualitative)
Less than 0.1	Excellent
Between 0.1 and 0.2	Very good
Between 0.2 and 0.4	Good
Between 0.4 and 0.8	Fair
Between 0.8 and 1.6	Poor
Greater than 1.6	Very poor

- g) Derivatives to be computed for FDM calculation:
- The first order derivatives of the Lo and Hi datasets are to be calculated.
 - The second order derivatives of the Hi datasets are to be calculated.
 - A simple difference method is used to generate the first and second derivatives $Lo'(i) = Lo(i+1) - Lo(i-1)$. A correction is applied at the domain's boundaries:

For the left edge of the domain:

$$\frac{dLo}{dx}(i) = \frac{Lo(i+1)-Lo(i)}{\Delta} \quad (15)$$

For the right edge of the domain:

$$\frac{dLo}{dx}(i) = \frac{Lo(i)-Lo(i-1)}{\Delta} \quad (16)$$

For all the internal points:

$$\frac{dLo}{dx}(i) = \frac{Lo(i+1)-Lo(i-1)}{\Delta} \quad (17)$$

Where Δ is set to 1 in all cases.

h) Computation of point-by-point FDM:

$$FDM_1(f) = \frac{|Lo_1'(f)| - |Lo_2'(f)|}{\frac{2}{N} \sum_{i=1}^N (|Lo_1'(i)| + |Lo_2'(i)|)} \quad (18)$$

$$FDM_2(f) = \frac{|Hi_1'(f)| - |Hi_2'(f)|}{\frac{6}{N} \sum_{i=1}^N (|Hi_1'(i)| + |Hi_2'(i)|)} \quad (19)$$

$$FDM_3(f) = \frac{|Hi_1''(f)| - |Hi_2''(f)|}{\frac{7.2}{N} \sum_{i=1}^N (|Hi_1''(i)| + |Hi_2''(i)|)} \quad (20)$$

i) Calculate point-by-point value of FDM

$$FDM(f) = FDM_1(f) + FDM_2(f) + FDM_3(f) \quad (21)$$

j) Compute the confidence histogram for FDM, similar to the one created for ADM.

k) Implementation of Grade and Spread

The representative nature of FSV has been enhanced by weighting ADM and FDM in the GDM computation according to the following [20], as opposed to deeming them equally essential.

- The degree to which the histogram bars are concentrated at either extreme (Grade)
- The degree to which the histogram bars are distributed around the mean value (Spread)

A user-specified amount (referred to as the "threshold" and by default set at 85%) of the total samples of the data sets that need to be compared is factored in when calculating the GRADE score. The classes range from the best (Excellent) to the worst (Very Poor). There are six possible GRADE values: 1 being the greatest quality and 6 being the lowest. The GRADE range is the range of the courses that are considered when calculating the GRADE value.

The SPREAD is determined by counting the number of classes, sorted from most populous to least popular, that contain a user-defined portion (called a "threshold"), which is by default set at 85% of the total samples of the data sets to be compared.

Now, computing the weights of K_{ADM} & K_{FDM} , the algorithm is [20]:

```

If  $Spread_{ADM} < Spread_{FDM}$ 
Then
     $K_{ADM} = 1$ 
     $K_{FDM} = Spread_{ADM} / Spread_{FDM}$ 
Else if  $Spread_{ADM} > Spread_{FDM}$ 
Then
     $K_{FDM} = 1$ 
     $K_{ADM} = Spread_{FDM} / Spread_{ADM}$ 
Else
     $K_{ADM} = 1$ 
     $K_{FDM} = 1$ 

```

1) Computation of Global Difference Measure (GDM):

$$GDM = \sum_{fmin}^{fmax} \sqrt{k_{ADM} \cdot (ADM(x))^2 + k_{FDM} \cdot (FDM(x))^2} \quad (22)$$

i) Now, A similar confidence interval for GDM and a Qualitative single value of fitness can be obtained from the GDM values.

This Global Difference Measure value provides us with an overall comparison of any two datasets.

5. Comparison of different models

A comparative analysis is being conducted to evaluate the performance of various candidate surrogate models. This includes models such as Wide Band Kriging, those provided by UQLab [4,5] (e.g., Kriging, Polynomial Chaos Expansion), Deep Infinite Mixture Gaussian Process (DIM-GP) offered by OptisLang, and others various Machine Learning models. Through this comparison, the research strives to achieve a balance between computational efficiency and model accuracy. The process involves fine-tuning these models to optimize the trade-off between computation time and relative error.

3.3 Conclusion:

This research investigated the application of surrogate models for achieving electromagnetic compatibility (EMC) in power electronic circuits. Building upon existing high-fidelity models, the focus shifted towards exploring faster and less computationally demanding alternatives. Various mathematical models, implementable with software were investigated.

The analysis centered on filter and DC-DC converter circuits, with emphasis on developing a suitable model for the latter. Validation techniques like Feature Selective Validation ensured model generalizability. A comparative analysis of different surrogate models, including Wide Band Kriging and Deep Infinite Mixture Gaussian Process has been performed. This research aims to achieve an optimal balance between computational efficiency and model accuracy for faster and more streamlined EMC compliance assessment in power electronic circuits.

The section further talks about the implementation of Feature Selective Validation, an important validation method devised to provide a qualitative value to the comparison of any two datasets. FSV works well when the datasets are visually similar to the human eyes and hence standard numerical error fails to give us a trustworthy Error or comparison value.

CHAPTER 4

RESULT ANALYSIS

4.1 Introduction:

This section talks about the analysis of the results obtained by me till now. It is about analysing the variation of hyperparameters of the Kriging model, followed by understanding the model's accuracy with respect to change of training points, benchmarking of Ansys OptisLang and the results from Feature Selective Validation.

4.2 Result Analysis:

1. Analysing how the hyperparameters of the Kriging model vary

Table 4.1 P values

Number of training points (n)	Cs	L	Csr	Lsr	EPC	Cs1	Lsr1
50	1.381	1.410	1.999	1.999	1.999	1.728	0.848
100	1.649	1.513	1.910	0.642	1.491	1.999	0.675
200	1.421	0.903	1.721	1.999	1.427	1.999	0.849
400	0.888	0.959	1.669	1.322	1.323	1.863	0.749

Table 4.2 θ values

Number of training points (n)	Cs	L	Csr	Lsr	EPC	Cs1	Lsr1
50	1.619	1.176	5.794	0.065	0.285	0.364	1.195
100	1.604	1.279	3.383	0.235	0.422	0.450	0.954
200	1.248	0.968	3.544	0.258	0.459	0.517	0.913
400	1.128	1.038	2.790	0.281	0.498	0.486	0.767

Here P and θ are the hyperparameters, p being the trend parameter, representing the deterministic part of the model, basically captures the large scale variations whereas θ is the spatial correlation parameter, it defines the range of the spatial dependencies between data points; with n being the number of training points given to the model.

On seeing θ table,

for n=50 : $Csr > Cs > Lsr1 \geq L > Cs1 > EPC > Lsr$

for n=100: $Csr > Cs > L > Lsr1 > Cs1 > EPC > Lsr$

for n=200: $Csr > Cs > L > Lsr1 > Cs1 > EPC > Lsr$

for n=400: $Csr > Cs > L > Lsr1 > Cs1 \leq EPC > Lsr$

So, we could say that C_{sr} is the most important or most affecting parameter. Followed by C_s and L_{sr1} . This Conclusion would be significant for the research as when we are to begin with the optimisation process, we will already have some idea as to how to tweak the components.

Table 4.3 Components of the circuit

Name of the Component	Function of the component
L	Common mode choke inductance
C_{sr}	Common mode parasitic capacitance
EPC	Parallel Capacitance of CM choke inductance
C_s	Capacitance-Input side
L_{sr}	Parasitic Inductance (input)
C_{s1}	Capacitance-Output side
L_{sr1}	Parasitic Inductance (output)

2. Model Accuracy with change in the number of trainings points.

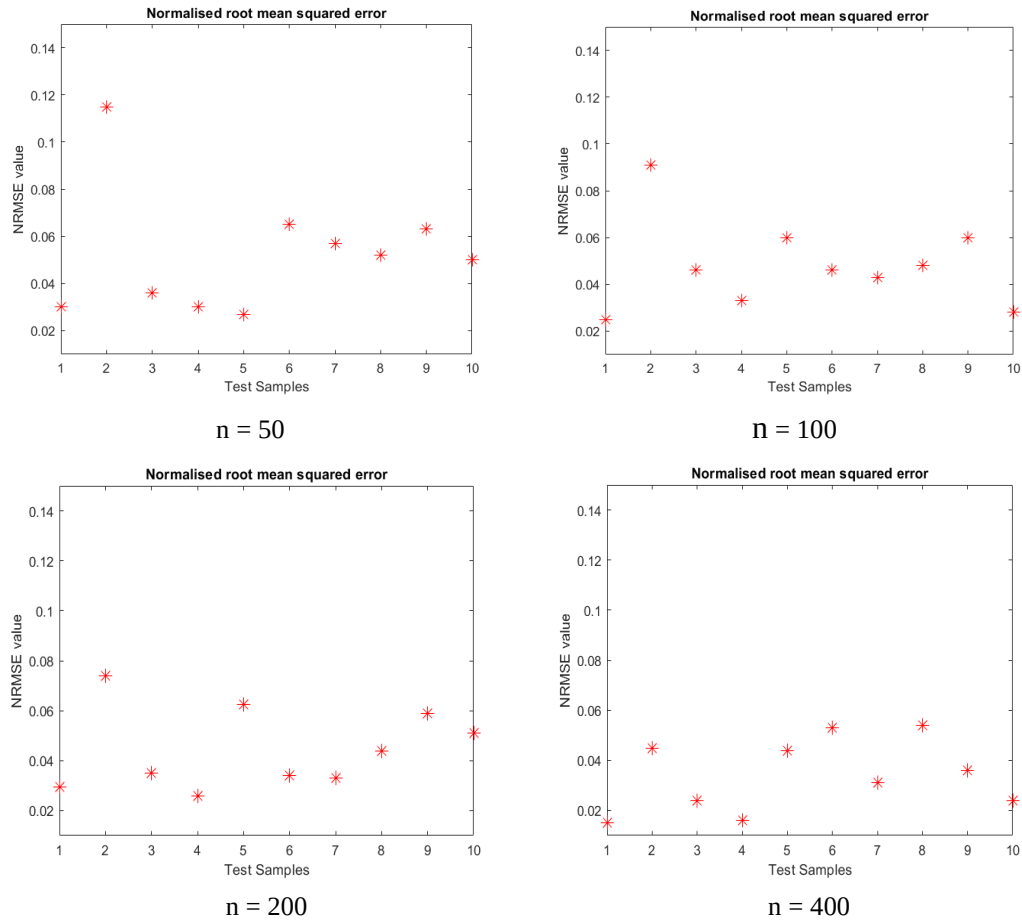


Figure 4.1: Comparing the NRMS error for different number of training points

The X-axis corresponds to the frequency range and Y-axis corresponds to the Insertion Loss (in decibels) measured for the Filter. It basically tells us the amount of EMI measured.

Mean of Error values: $n=50 \Rightarrow 5.25\%$, $n=100 \Rightarrow 4.80\%$, $n=200 \Rightarrow 4.44\%$, $n=400 \Rightarrow 3.39\%$
(The error was measured using Normalised Root mean square method)

Some conclusions that could be derived from here was:

- Error drops as the number of training sample increases.
- The increase of training samples improves the model more so in the region with sharper resonances.

3. *Improvement on computation time by altering the theta function.*

The existing code base developed by colleagues was modified, resulting in a reduction in computation time. The time taken for model generation improves, from 10540.708765 seconds to 7095.090947 second (for a model training on 102 frequency points and 400 training points)

This is important as it enhances efficiency, allowing for faster iterations and quicker feedback during the model development process. This can be particularly beneficial when dealing with large datasets or needing to perform multiple models runs, as it reduces overall time expenditure. Also, improved computational efficiency translates to lower computational costs and resource usage, making the process more sustainable and scalable.

In the context of kriging, where model accuracy and precision are paramount, reducing computation time without compromising these qualities can lead to more effective utilization of resources and potentially enable the handling of more complex or larger datasets within practical time frames.

4. *Analysis of Wide Band Kriging model (with respect to number of training points and computation time)*

A balance between the computational time required by the surrogate models and the associated error was sought in this investigation.

Legends: The yellow circle represents the Error percentage (smaller the circle, better our model's prediction) and the red cross represents the Computation time (smaller the cross, faster is the model). X axis corresponds to number of training samples and Y axis corresponds to the number of frequency points for each sample.

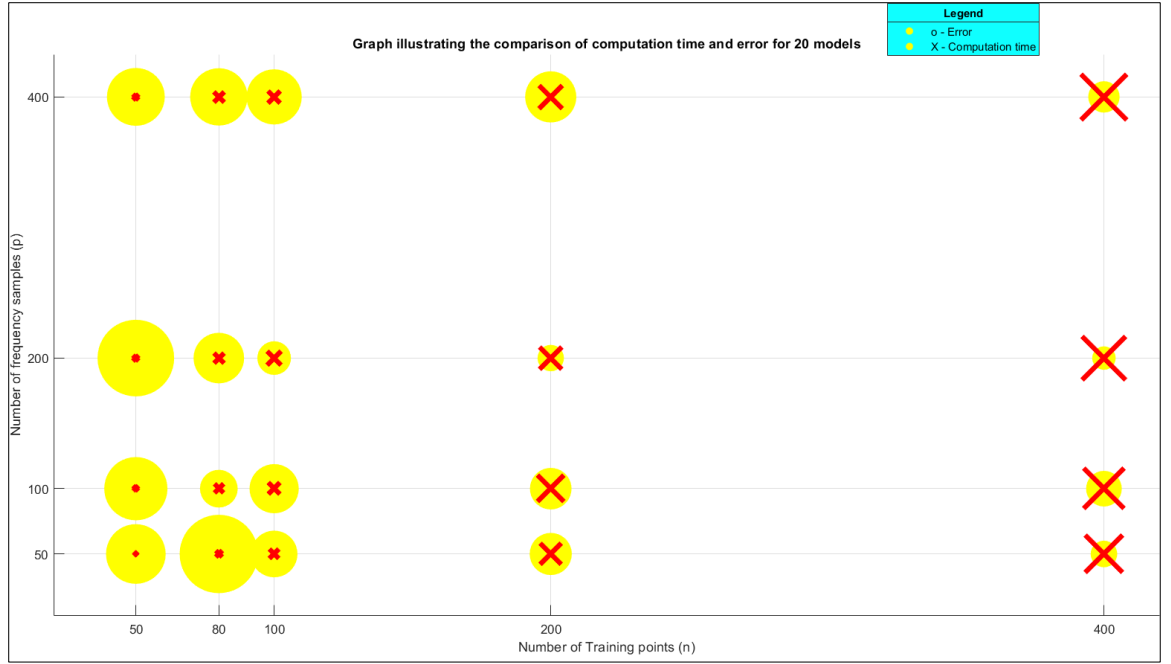


Figure. 4.2: Comparison of computation time and error

This pictorial representation of the Costs of the training models, that are the training time and error helps us to decide which model is suitable for further usage.

Some Conclusions derived from this comparison were:

- Computation time increases with the increase in ‘n’ or the number of training points and isn’t significantly affected by increasing number of frequency points.
- A reason for this could be that the main time-consuming part of the model building is the optimization of the p and theta matrix, which in turn depends upon the number of training points, and the frequency doesn’t come into the picture until the part of building the surrogate model.
- Error isn’t significantly affected by just changing number of frequency points.
- There is a saturation with respect to the accuracy and the number of training points (also frequency samples). To overcome this, we need a better sampling method, so that the region in the curves with bad prediction (the sharp resonances) also improves.

5. *Benchmarking of Deep Infinite Mixture Gaussian Process (DIM-GP) model from OptisLang*

DIMGP model from OptisLang was used to create a surrogate model for the 7D (7 varying components) Filter circuit talked about in the previous sections.

The results obtained from the model were:

(Testing was done on samples generated from Random Latin Hypercube for 100 testing samples)

Table 4.4 Comparison of OptisLang model trained on different number of points

Number of training points (n)	Computation time	Error % (NRMSE)
100	1625 seconds	5.1%
200	2500 seconds	4.6%
400	4168 seconds	4.1%

Some conclusions that could be derived were:

- The resonances, which are generally tough for other models to capture, were also captured very well by DIMGP.
- The Model computation time is significantly less when compared to Wide Band Kriging.

Hence, we could conclude that OptisLang DIMGP model works very well.

6. Implementation of Feature Selective Validation in Python

The FSV tool has been successfully implemented in Python. Feature Selective Validation (FSV) is a technique used to compare data from simulations and experiments in electromagnetic compatibility (EMC) applications [17]. It goes beyond a simple "goodness-of-fit" by analysing specific features and providing both numerical and qualitative assessment for improved validation [20].

A working example of FSV has been shown here:

The following figure depicts two very similar datasets plotted as two curves.

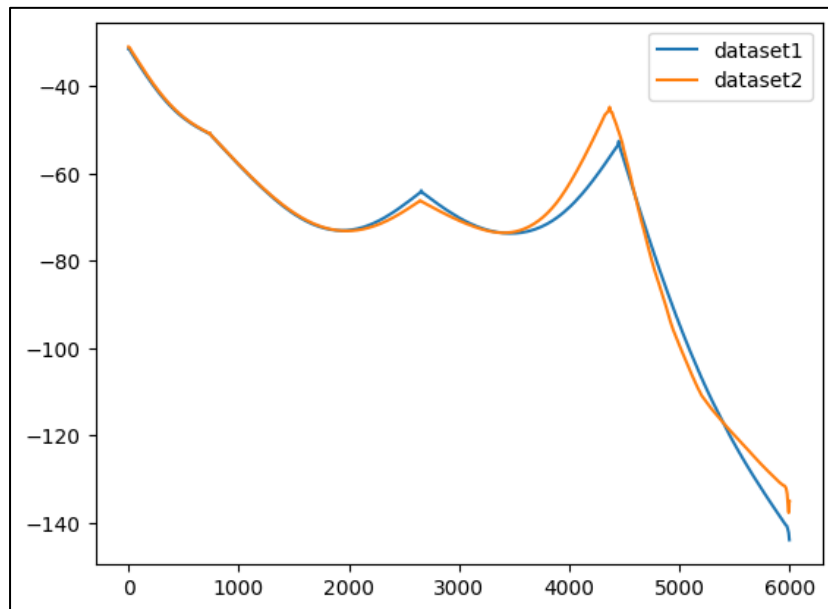


Figure. 4.3 Two datasets to be compared

(Both datasets consist of 6000 points across the range.)

On Calling the Function FSV for the shown datasets, the following Figures were obtained

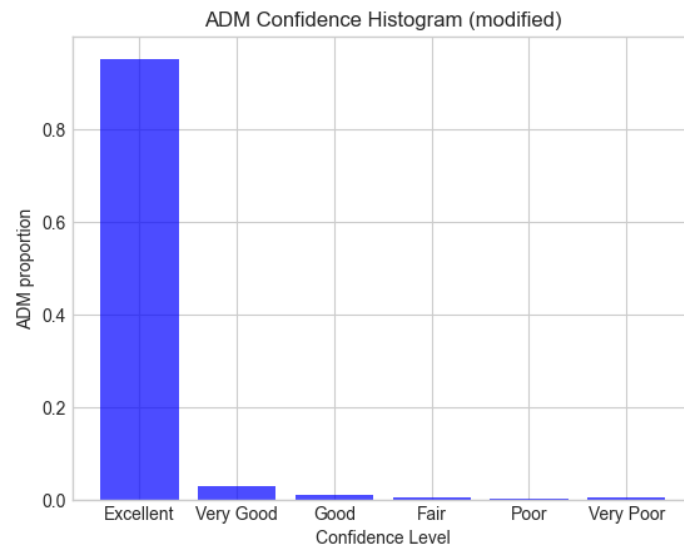


Figure. 4.4 Confidence Interval of Amplitude Difference Measure

The inference from Figure 6 is that the overall ‘trend’ of the two datasets is Excellent (highly matching) for around 90% of the dataset points of the two curves. Hence the similarity is very high.

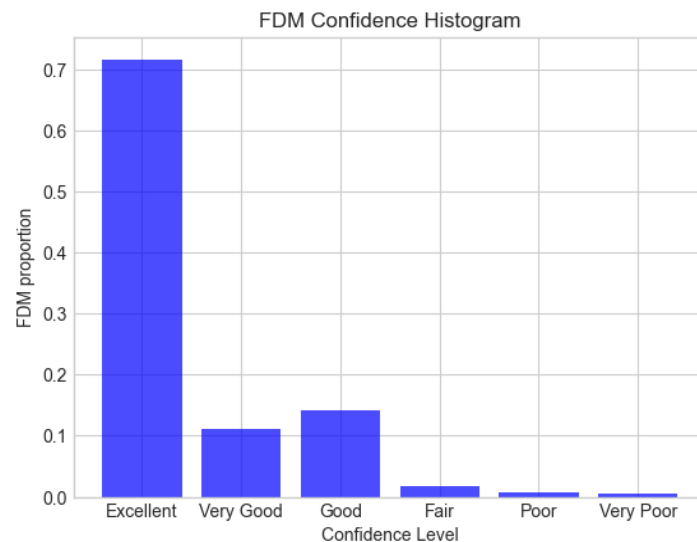


Figure. 4.5 Confidence Interval of Feature Difference Measure

The inference from Figure 7 is that the fine features of the two datasets are Excellent (highly matching) for close to 70% the dataset points of the two curves.

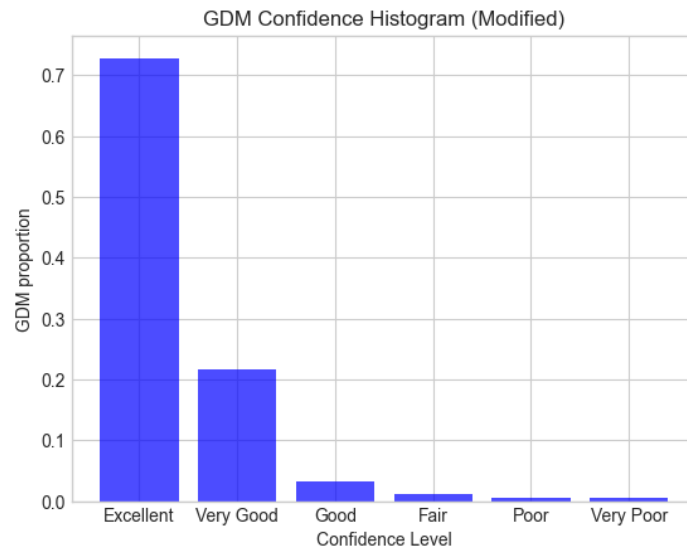


Figure. 4.6 Confidence Interval of Global Difference Measure

The inference from Figure 8 is that the Two curves are highly alike each other as close to 70% of the comparison of the datapoints lie in the Excellent range (highly matching).

The Single Value of the various comparisons obtained are:

Comparison type	Quantitative Value	Qualitative Value
ADM	0.079	Excellent
FDM	0.111	Very good
GDM	0.121	Very good

The One-values for ADM, FDM, GDM respectively tells us that the overall ‘trend’ of the two curves is matching in an excellent manner, the finer features (the resonances) have very slight differences and so the overall comparison (GDM) tells that the two datasets (or curves) are very identical.

Hence, from these results we can provide a Quantitative (Numerical) as well as a Qualitative value to the comparison of the datasets.

This is important as it overcomes many of the limitations of statistical approaches by relating the results of the comparison to the original data on a point-by-point basis (in addition to relating the results to natural language descriptions).

4.3 CONCLUSIONS:

Up to this point, surrogate models in our possession have been subjected to a process of improvement, comparison, and modification. While some models exhibited better performance than anticipated, none displayed significant deviations between predicted and actual results. The FSV implemented in python works well and up to the expectations.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE OF WORK

5.1 Conclusion:

In this work, the application of surrogate models for achieving electromagnetic compatibility (EMC) in power electronic circuits was investigated. By leveraging existing high-fidelity models, the study explored faster and less computationally demanding alternatives to improve the efficiency of EMC assessments.

The analysis primarily focused on filter and DC-DC converter circuits, aiming to develop suitable surrogate models. Key validation techniques, such as Feature Selective Validation (FSV), were employed to ensure the generalizability and reliability of these models. FSV provided a robust method for qualitative comparison by integrating amplitude and feature-based measures, offering a more comprehensive error analysis compared to standard numerical methods.

The research also included a comparative analysis of various surrogate models, including Wide Band Kriging, and Deep Infinite Mixture Gaussian Process (DIM-GP). This comparison aimed to balance computational efficiency and model accuracy, essential for streamlined EMC compliance assessments.

The results demonstrated that surrogate models could effectively reduce computation time without significantly compromising accuracy. The study highlighted the importance of adaptive sampling in optimizing data collection and improving model performance. Furthermore, it underscored the potential of these models to enhance decision-making capabilities in complex systems with high-dimensional input spaces.

Future work should focus on refining these surrogate models further and exploring additional techniques to enhance their accuracy and efficiency. Implementing adaptive sampling strategies and investigating other machine learning approaches could provide valuable insights and improvements.

Overall, this research lays a strong foundation for advancing EMC optimization techniques, contributing to more reliable and efficient power electronic systems.

5.2 Future Scope of work:

- **Implementing Adaptive Sampling:**
Adaptive sampling in metamodeling optimizes data collection by strategically selecting sample points based on the evolving understanding of the system's behaviour [9,11,12]. This approach minimizes the number of evaluations needed to construct accurate surrogate models, making it efficient for complex systems with high-dimensional input spaces. By focusing on informative regions, adaptive sampling accelerates design optimization and enhances decision-making capabilities.
- **Beginning with the Optimisation of the circuits:**
In the near future, we begin applying the most suitable Surrogate model we have on our hands to optimise the circuits, such that they are within the CISPR-25 limits [13].
(The international standard CISPR 25 specifies guidelines and limitations for measuring radio disturbances in the 150 kHz to 5 925 MHz frequency range. Any electrical or electronic component meant for use in cars, trailers, or other devices must adhere to the standard.)
- **Implementation of Physics Based metamodeling:**
Physics-based surrogate modelling for Electromagnetic Compatibility (EMC) integrates electromagnetic field theory with machine learning techniques to construct efficient approximations of complex EMC phenomena. By incorporating the underlying physics principles governing electromagnetic interactions into the modelling process, these surrogate models offer more accurate predictions and deeper insights into EMC performance. This approach will enable us to simulate EMC behaviour with reduced computational resources, aiding in the design and optimization of electronic systems to mitigate electromagnetic interference and ensure compliance with EMC standards.

CHAPTER 6

HEALTH, SAFETY, RISK AND ENVIRONMENT ASPECTS

6.1 *Health and Safety Considerations*

During the execution of this project, several health and safety considerations were prioritized to ensure the well-being of personnel and mitigate potential risks associated with experimental setups and operations:

1. Workplace Safety: Strict adherence to safety protocols while in the working environment.
2. Laboratory Safety: Maintaining a safe laboratory environment for conducting experiments with hardware components, ensuring compliance with safety standards and guidelines.
3. Emergency Procedures: In-place protocols for equipment failures, Fire alarm.

6.2 *Risk Management*

Effective risk management strategies were implemented throughout the project to identify, assess, and mitigate potential risks associated with research activities:

1. Risk Assessment: Carrying out thorough risk analyses to pinpoint any dangers associated with experimental configurations and data gathering techniques.
2. Data Security: Implementing protocols to protect sensitive data collected during experiments, new results obtained, including secure storage and access control measures.
3. Compliance and Ethics: Upholding ethical standards in research conduct, including participation consent, data privacy, and compliance with the institutional guidelines and regulations.

6.3 *Environmental Aspects*

Environmental considerations were integrated into the project to minimize environmental impact and promote sustainable practices:

1. Energy Efficiency: Promoting energy-efficient practices in hardware configurations and computational workflows to reduce the environmental footprint associated with intensive computing tasks.

2. **Waste Management:** Adhering to waste management guidelines for the disposal of electronic waste, packaging materials, and other consumables used in experimental setups and office workspace.
3. **Resource Consumption:** Considering the energy requirements of large machine learning models and GPUs used for training, the workspace energy requirement, efforts were made to optimize energy use and reduce overall resource consumption.

By integrating these health, safety, risk management, and environmental considerations into the project's execution, we aimed to conduct research responsibly while advancing our research knowledge.

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Expected date of completion of project	30 th June 2024		
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